Explain the TCP three-way handshake and teardown, with a table of TCP flags (SYN, ACK, FIN, RST, PSH, URG)

TCP (Transmission Control Protocol) is a connection-oriented, reliable transport-layer protocol used to establish communication between devices over a network.

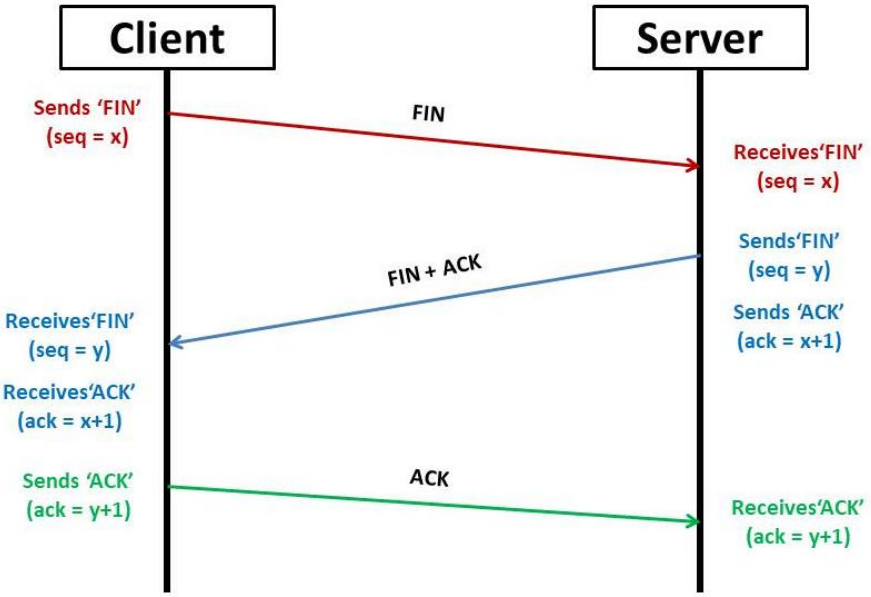
TCP operates at the Transport Layer (Layer 4) of the OSI model. It ensures that data is delivered accurately, in the correct sequence, and without unnecessary duplication.

The main features of TCP include:

- Connection-oriented communication: Establishes a connection before transmitting application data.
- Reliable delivery: Uses acknowledgments and retransmissions to recover from packet loss.
- Sequencing: Maintains the correct order of transmitted data.
- Flow control: Prevents a sender from overwhelming the receiver.
- Congestion control: Adjusts the transmission rate according to network conditions.
- Error detection: Uses checksums to detect corrupted TCP segments.

1. TCP Three-Way Handshake

The TCP three-way handshake is the process used to establish a connection between a client and a server before data transmission.



Step 1: SYN (Synchronize)
The client initiates communication by sending a TCP segment with the SYN flag enabled.

- The client selects an Initial Sequence Number (ISN).
- The SYN flag is set to 1.
- The client enters the SYN-SENT state.
- The SYN segment may also advertise connection parameters such as the Maximum Segment Size (MSS).

Example:

SYN, Seq=100

Step 2: SYN-ACK (Synchronize-Acknowledgment)
The server responds with a SYN-ACK segment.

- Both SYN and ACK flags are set.
- The server acknowledges the client's initial sequence number.
- The server generates its own initial sequence number.
- The server enters the SYN-RECEIVED state.

Example:

SYN, ACK, Seq=400, Ack=101

Step 3: ACK (Acknowledgment)
The client sends an ACK segment confirming the server's response.

- The ACK flag is set.
- The client acknowledges the server's initial sequence number.
- The client enters the ESTABLISHED state.
- The server enters ESTABLISHED after receiving the acknowledgment.

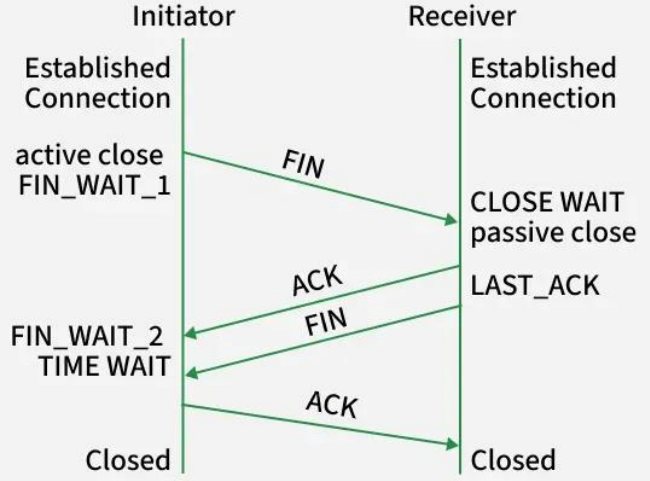
Example:

ACK, Seq=101, Ack=401

The TCP connection is now established and application data can be exchanged…

2. TCP Connection Teardown

TCP uses a graceful connection termination process, commonly involving four segments.
Typical four-segment TCP connection termination



Step 1 - FIN: The client sends a FIN segment to indicate that it has finished sending data and enters the FIN-WAIT-1 state.

Step 2 - ACK: The server acknowledges the FIN and enters CLOSE-WAIT. After receiving the acknowledgment, the client enters FIN-WAIT-2.

Step 3 - FIN: When the server finishes sending its remaining data, it sends its own FIN and enters LAST-ACK.

Step 4 - ACK: The client acknowledges the server's FIN and enters TIME-WAIT. The server closes after receiving this acknowledgment. The client closes after the TIME_WAIT period.
A connection may also be terminated abruptly using the RST flag.

3. TCP Flags

Flag	Full Form	Function
SYN	Synchronize	Initiates a TCP connection and synchronizes sequence numbers
ACK	Acknowledgment	Confirms receipt of data or control segments
FIN	Finish	Gracefully closes one direction of communication
RST	Reset	Immediately resets or rejects a TCP connection
PSH	Push	Requests prompt delivery of received data to the application
URG	Urgent	Indicates that the urgent pointer is meaningful

4. Security Significance of TCP Flags

TCP flags are important in network security monitoring because unusual flag combinations or connection patterns may indicate malicious activity.

- SYN Flooding: Large numbers of SYN requests may exhaust a server's connection resources.
- SYN Scanning: SYN packets sent to multiple ports may indicate network reconnaissance.

- RST Scanning or Connection Resets: Frequent RST packets may indicate rejected connections, scanning, or network problems.
- FIN Scanning: FIN packets sent without an established connection may be used to probe TCP port behavior.
- NULL and Xmas Scans: Unusual TCP flag combinations may indicate specialized port-scanning techniques.